

TrES-5b

R-0950

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_230624 · created 2026-07-29T14:21:33+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460119.7345 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.022
Transit depth (Rp/Rs)^2	2.29 +/- 0.66 [%]
Orbital Inclination (inc)	85.95 +/- 2.19
Airmass coefficient 1 (a1)	1.043 +/- 0.018
Airmass coefficient 2 (a2)	-0.027 +/- 0.016
Scatter in the residuals of the lightcurve fit is	1.76 %
Best Comparison Star	#1 - [276, 377]
Optimal Method	PSF photometry
Transit Duration (day)	0.0788 +/- 0.0093

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.151 $\pm$ 0.022 vs 0.142 (deviation 0.41 σ)
Duration fit vs published	0.0788 d vs 0.0758 d (deviation 0.32 σ)
Mid-time O-C	-9.7 min
Depth SNR	6.9
Residual scatter	1.76 %

Light curve

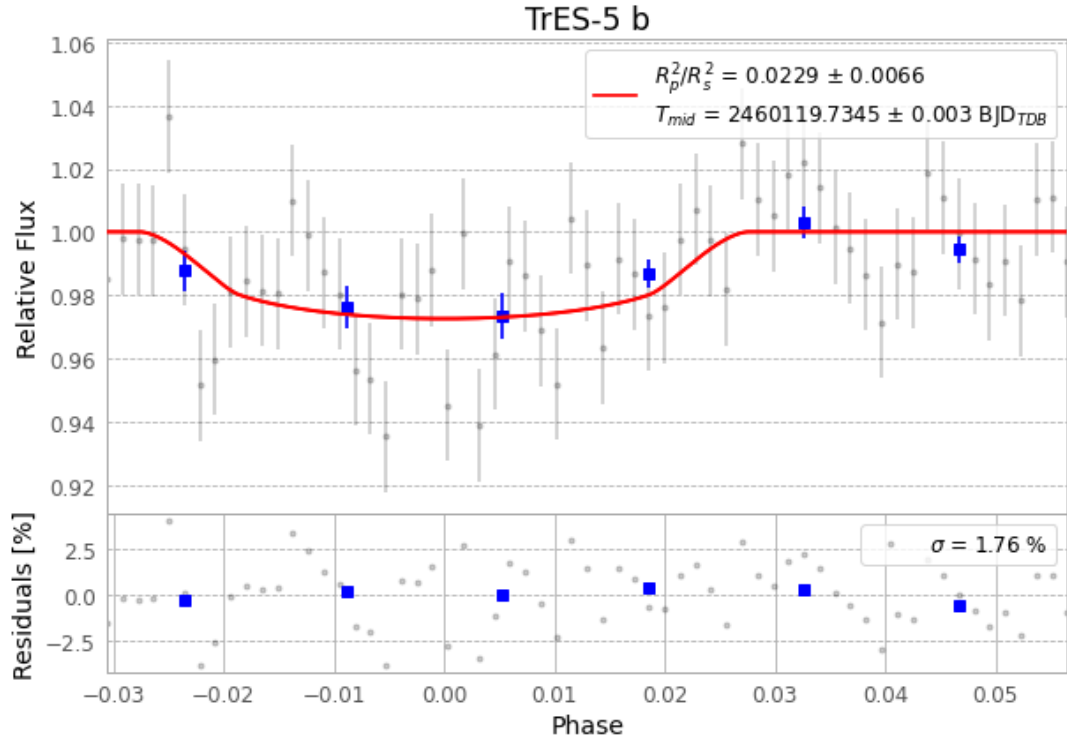


Figure 1 — FinalLightCurve_TrES-5 b_2023-06-23.png (SHA-256 28aa3efbea0216ea...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [293, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9b8c2f7e81d3afb5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

63 FITS frames (41 MB), TRES-5230624043016.FITS → TRES-5230624073615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-230624.log (a468669a7daa...), FinalLightCurve_TrES-5 b_2023-06-23.png (28aa3efbea02...), FinalLightCurve_TrES-5 b_2023-06-23.csv (a55dd72d0c4d...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 14:21Z.